



MODELLING DIFFUSIVE SIGNALS FOR THE GERMINATION OF *ASPERGILLUS* CONIDIA

THESIS DEFENCE

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July 24, 2025

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CONTENTS



1 Introduction

- Research outline
- Aspergillus* moulds
- Spore germination
- Research questions

2 Computational models

- Analytical models
- Numerical models

3 Experiments

- Comparative validation
- Inducer permeation
- Single-spore inhibition
- Adsorption
- Multi-spore inhibition
- Cluster experiments
- Germination models
- Parameter estimation

4 Conclusion

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- Numerical models

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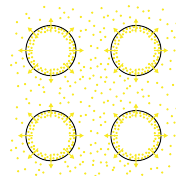
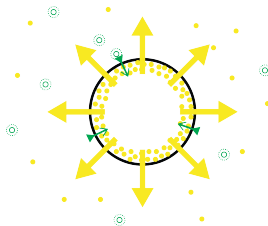
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RESEARCH OUTLINE



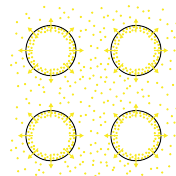
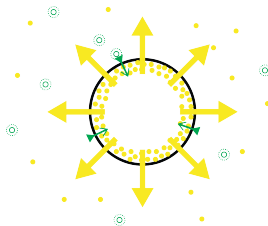
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- » Goal: a mathematical model for germination triggering in spores via passively diffusing **inducing** and **inhibitory** chemical signals.
- » Strong focus on an inhibitory **crowding effect**.
- » Validation through **parameter estimation** on germination data.



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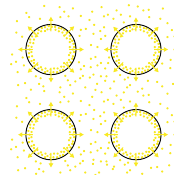
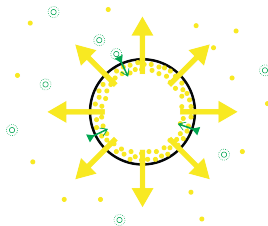
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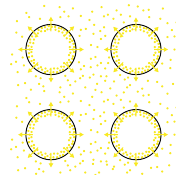
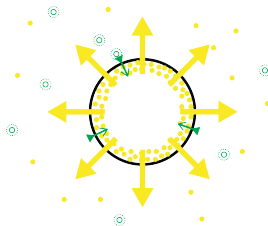
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ASPERGILLUS MOULDS



Plant/human pathogens and molecular factories

» Relevance to **medicine**, **food industry** and **circular economy**.

» Notable species:

- *A. niger*
- *A. nidulans*
- *A. flavus*
- *A. fumigatus*
- *A. oryzae*

» Reproduction through asexual spores (**conidia**)

- Dormant spores - highly resistant, partly due to **hydrophobin** coating
- Germlings and mycelium - less resistant, restructured cell wall



Figure: *A. niger*

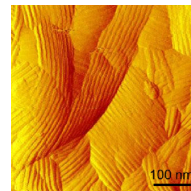


Figure: Hydrophobins [6]



Figure: Bioreactor

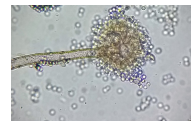


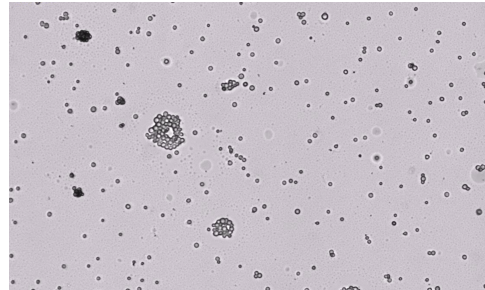
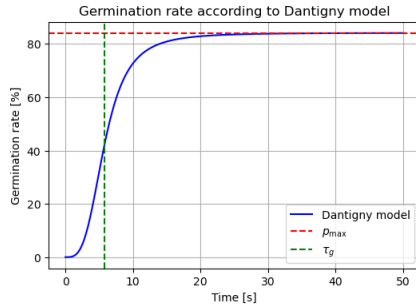
Figure: Conidiophore

SPORE GERMINATION



Coordination and adaptivity

- » Germination typically begins within **4 hours** of cultivation.



SPORE GERMINATION



Inductive signalling

- » Carbon source molecules (glucose, amino acids) indicate the presence of nutrients in the medium.
- » Carbon sources bind to **G-proteins** and **RasA** proteins embedded in the cell wall, activating dormancy break mechanisms such as the cAMP-PKA pathway.



Figure: D-glucose [PubChem]



Figure: Alanine [PubChem]

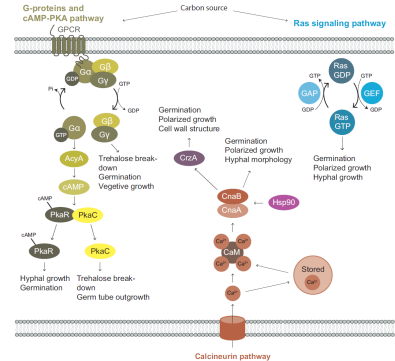


Figure: Carbon signalling pathways [1]

SPORE GERMINATION



Inhibition

- » When spores are cultivated at a **high density**, a **smaller percentage** of them germinates.
- » This is likely not due to a competition for resources, since spores **do not consume nutrients**.
- » More probably, spores release **inhibitory signals**, which saturate in dense spore cultures.

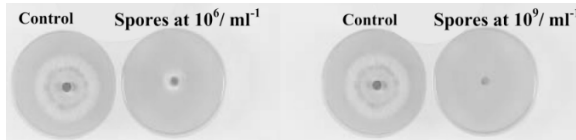


Figure: Auto-inhibition through spore density [2]

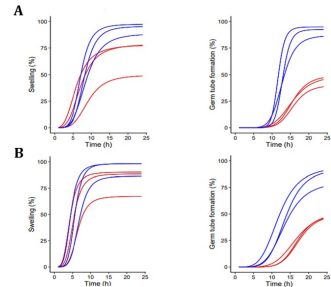


Figure: Time course of germination percentage (biological triplets) in a medium with alanine (A) and proline (B): 5000 spores (blue) and 40000 spores (red) per 150 μL [4].

SPORE GERMINATION



1-octen-3-ol

- » **Volatile organic compounds** (8-carbon oxylipins) are found in the headspace of *Aspergillus* and *Penicillium* cultures.
 - Relatively small molecules;
 - Heat-stable;
 - Known to participate in fungus-host communication.
- » Of these, **1-octen-3-ol** is identified as a prominent germination inhibitor (when added externally).
- » Other potential inhibitors (in *A. nidulans*) are 3-octanone and 3-octanol.
- » Controversial role in germination.
- » If the inhibitor is **not continuously replenished** in a dormant spore, does its **passive depletion enable germination?**

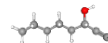


Figure: 1-octen-3-ol [PubChem]

RESEARCH QUESTIONS



Are signals timed by diffusion/permeation?

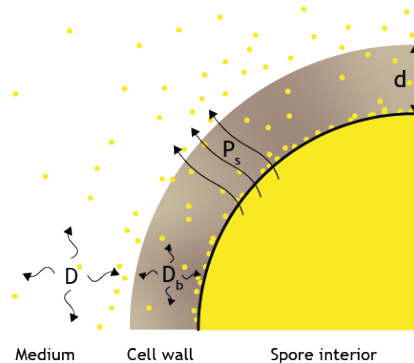
- » The flux J of molecules through a homogeneous medium is characterised by a **diffusion constant** D :

$$J = -D \frac{\partial c}{\partial x}. \quad (1)$$

- » The passage of molecules through a barrier is determined by a **permeation constant** $P_s \sim D/d$:

$$J = -P_s \Delta c. \quad (2)$$

- » In both cases, the difference in the **concentration** c is driving the flux.
- » Do known diffusivities/permeabilities/concentrations lead to signal timings consistent with germination?



RESEARCH QUESTIONS



Is diffusion/permeation modulated by (de)saturation?

- » Inhibitor molecules diffuse as time-dependent Gaussian distributions around the sources.
- » A denser array of sources causes inhibitor **saturation** in the extracellular medium.
- » When the concentration in the medium is **similar** to the concentration in the spore, the molecules diffuse **slowly**.
- » Can **increased spore densities** change germination behaviour via external **inhibitor saturation**?

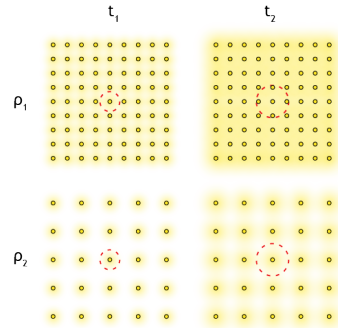


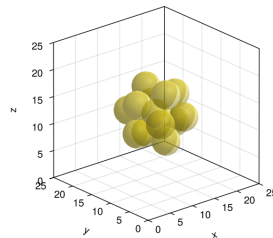
Figure: Saturation at different times and spore densities.

RESEARCH QUESTIONS



Obstructing effect of spore clusters

- » Spore volumes present **obstacles** for diffusion.
- » Spores surrounded by neighbours may **retain** more unreleased inhibitor.
- » Do spores inside aggregated clusters have a smaller chance for germination?



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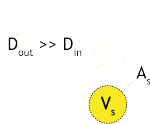
4 Conclusion

ANALYTICAL MODELS

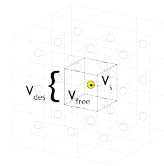


Simple permeation vs. volume-based model

$$c_{\text{in}}(t) = c_0 e^{-t/\tau} \quad (3)$$



$$c_{\text{in}}(t) = \phi c_0 + (1 - \phi) c_0 e^{-\frac{t}{\tau(1-\phi)}} \quad (4)$$



- » Homogeneous internal distribution
- » Fast external equilibration

- » Equal designated volume per spore
- » Volume fraction $\phi = \rho_s V_s$

NUMERICAL MODELS



Model resolutions

» Spore representation on macroscopic, spore and cell-wall scale

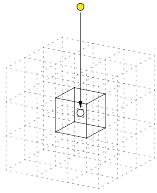


Figure: Low resolution.

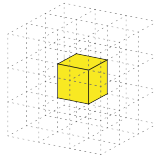


Figure: Medium resolution.

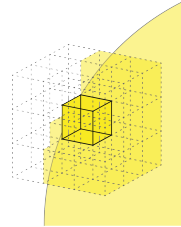


Figure: High resolution.

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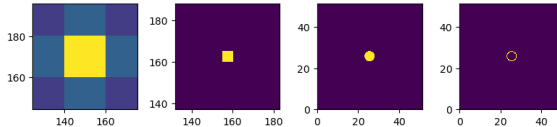
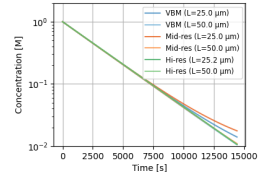
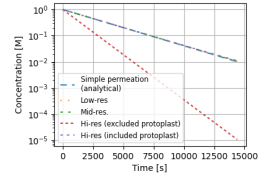
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COMPARATIVE VALIDATION



Consistency across scales

- » All models assuming **homogeneous inhibitor distribution** inside the spore yield the **same depletion trend**.
- » A high-resolution model with the same inhibitor quantity concentrated **only in the cell wall** yields a **faster depletion**.
- » The **medium-resolution** numerical model and the **volume-based** analytical model exhibit **saturation** at time scale of germination (around $t = 4$ h).

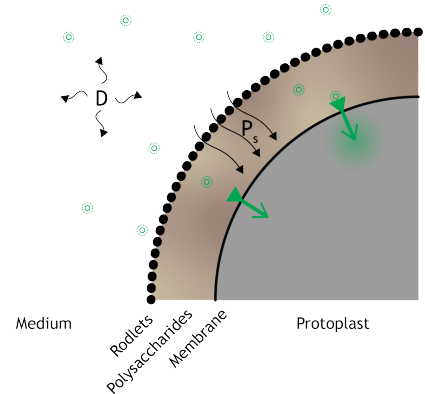


INDUCER PERMEATION



Role of the hydrophobin barrier

- » Glucose is **polar** and should permeate slowly through the outer **hydrophobin rodlet layer** of the spore.
- » Calculations show that glucose concentrations are still not equilibrated across the hydrophobin layer at $t = 4$ h.
- » Therefore, **cell wall permeability modulates inductive signalling**.



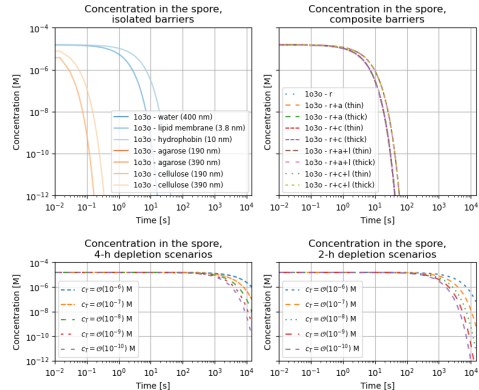
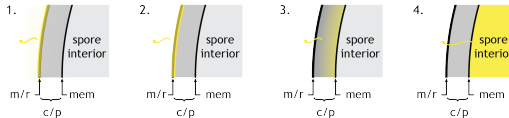
SINGLE-SPORE INHIBITION



Time scale inconsistency

- » 1-octen-3-ol is **less polar** and passes through the rodlet layer more easily.
- » Considering
 - different initial distributions of 1-octen-3-ol,
 - different cell wall consistencies (agarose-like to cellulose-like),

1-octen-3-ol **depletes within seconds to minutes**, not hours.



ADSORPTION



Permeation reduction through strong interactions

- » Adsorption in the **polysaccharide matrix** of the cell wall could explain the delayed release.
- » Langmuir adsorption parameters of yeast β -glucans indicate only a halving of P_s .

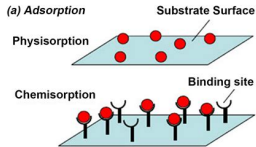


Figure: Types of adsorption [5].

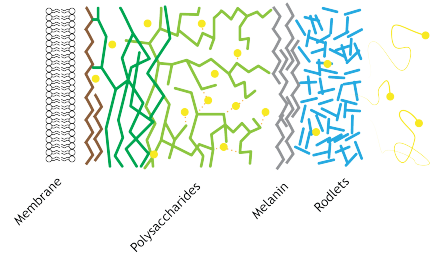


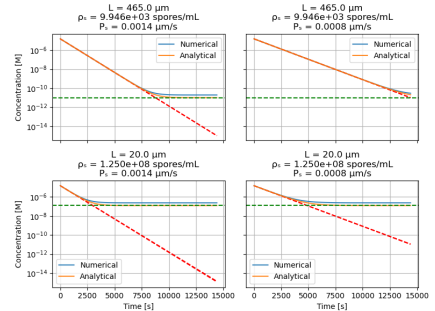
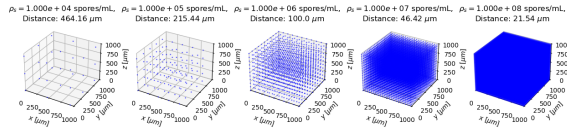
Figure: Adapted from [3].

MULTI-SPORE INHIBITION



Effect of global spore density

- » The spore number density ρ_s determines the **level of inhibitor saturation** in the spore.
- » It also mildly influences the **rate of inhibitor depletion** at early times.

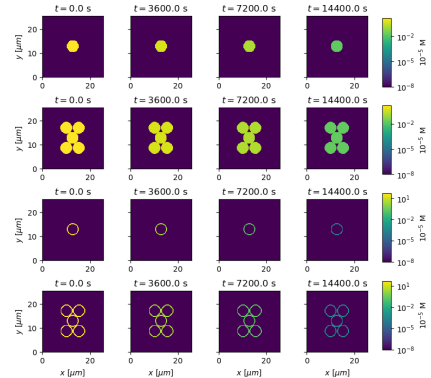


CLUSTER EXPERIMENTS



Effect of local spore density

- » A spore inside a dense cluster releases its inhibitor **only slightly slower** than a solitary spore.
- » This happens only when there is **direct contact** between spores (area of permeation is decreased).
- » Fast diffusion in the cluster cavities "rinses" the inhibitor away.

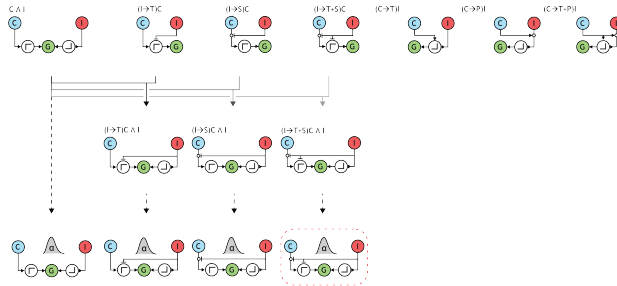


GERMINATION MODELS



Inducer-inhibitor interactions

- » Germination occurs when the inducing signal goes over and the inhibitory signal goes under a threshold. Induction and inhibition may influence each other in different ways.

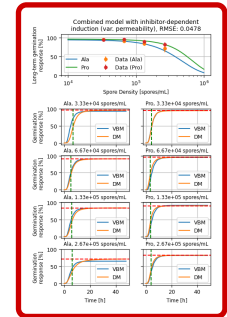
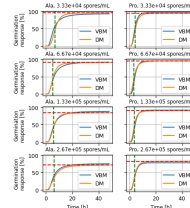
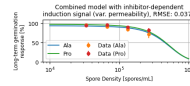
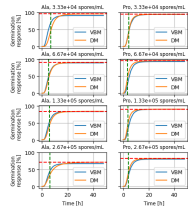
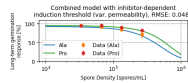


PARAMETER ESTIMATION



A. niger data

- » Models fit to time-dependent *A. niger* germination data with varying ρ_s .
- » In the best-fitting model, germination is **triggered by induction and by release of inhibition**, and
- » the **inhibitor modulates the inducing signal and its threshold**.

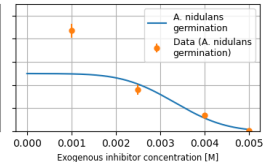
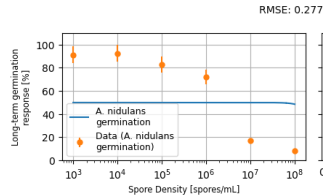
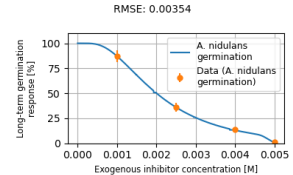
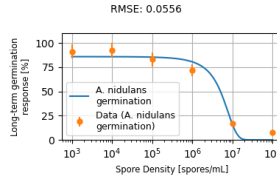


PARAMETER ESTIMATION



A. nidulans data

- » Best models also fit to steady-state *A. nidulans* germination data with
 - varying ρ_s ;
 - varying concentration of externally added 1-octen-3-ol.
- » Good separate fits, but **no parameter consensus** for both cases taken together.
- » Implies that **endogenous inhibition works differently** from the exogenous addition of 1-octen-3-ol.



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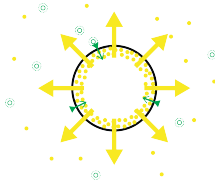
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CONCLUSION



» Diffusivity/permeability

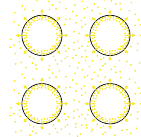
- Inductive signals are likely **limited by hydrophobin permeability**.
- Known cell wall and inhibitor properties lead to an **underestimation of the inhibitor release time** (possible causes: lag, adsorption, continuous synthesis, ...)



» Crowding

- **Global spore density** affects inhibitor depletion.
- **Local spore density** (clusters) - much less so.

» 1-octen-3-ol likely not alone in inhibiting germination.



THANK YOU!



BIBLIOGRAPHY I



- [1] Tim J. H. Baltussen et al. "Molecular Mechanisms of Conidial Germination in *Aspergillus* spp". In: *Microbiology and Molecular Biology Reviews* 84 (2019). URL: <https://api.semanticscholar.org/CorpusID:208642285>.
- [2] Gilma Silva Chitarra. *Germination inhibitors of fungal spores: Identification and mode of action*. Wageningen University and Research, 2003.
- [3] Neil A. R. Gow, Jean Paul Latgé, and Carol A. Munro. "The Fungal Cell Wall: Structure, Biosynthesis, and Function". In: *Microbiology Spectrum* 5 (2017). URL: <https://doi.org/10.1128/microbiolspec.funk-0035-2016>.
- [4] Maryam Ijadpanahsaravi et al. "The impact of inter- and intra-species spore density on germination of the food spoilage fungus *Aspergillus niger*.". In: *International journal of food microbiology* 410 (2023), p. 110495. URL: <https://api.semanticscholar.org/CorpusID:265268197>.

BIBLIOGRAPHY II



- [5] Kalebaila K. Kennedy, Kenneth J. Maseka, and Misheck Mbulo. “Selected Adsorbents for Removal of Contaminants from Wastewater: Towards Engineering Clay Minerals”. In: *Open Journal of Applied Sciences* 8 (2018), pp. 355–369. URL: <https://api.semanticscholar.org/CorpusID:134131306>.
- [6] Sarah Sze Wah Wong et al. “Species-Specific Immunological Reactivities Depend on the Cell-Wall Organization of the Two Aspergillus, Aspergillus fumigatus and A. flavus”. In: *Frontiers in Cellular and Infection Microbiology* 11 (2021). URL: <https://api.semanticscholar.org/CorpusID:232051314>.